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RESEARCH ARTICLE

Application of XDraw Algorithm to Diffraction Modeling Using 3D Vector Building Data

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ABSTRACT

Wireless network planning and optimization involves radio propagation modeling, which requires detailed terrain and building height data. In the past, raster-based building rooftop height data extracted from building footprints has been used down to 1 m resolution. For numerical evaluation of diffraction loss in case of raster-based input data, fast approximate algorithms are used, such as R2, XDraw, and their various implementations. Vector-based 3D city models with a high level of detail have recently gained significant interest. However, existing vector-data analysis algorithms are slow, and efficient numerical methods remain scarce. This work aims to develop a fast diffraction algorithm tailored to high-accuracy 3D vector building data. In the proposed model, diffraction loss is computed in accordance with the ITU-R P.526 recommendation, using the XDraw approximation to enable fast numerical processing. The algorithm runs in $\mathcal{O}(N^2)$ time by analyzing propagation in rings starting at the antenna location and progressing toward the edges of the analysis area. The performance and accuracy of the diffraction prediction model are benchmarked against the traditional raster-based XDraw algorithm, with results calculated using the 3D BAG open dataset for the city of Amsterdam.

1 | Introduction

Mobile networks are moving toward higher-GHz frequency bands near the sub-millimeter-wave spectrum, however, lower frequencies are still widely exploited and carry most of the traffic. While 5G today centers on 3.5 GHz, future deployments will target the upper mid-band 7–24 GHz frequencies (Testolina, Polese, and Melodia 2024; Kang et al. 2024). Higher frequencies suffer severe diffraction-related losses in dense urban environments, so accurate diffraction loss predictions are necessary for wireless network planning and optimization. Most current tools use rasterized rooftop heights down to 1 m resolution (Colpaert et al. 2018) to enable fast approximations (e.g., XDraw and its variants (Zhang et al. 2021)); however, as detailed vector-based 3D city models become more common, processing them remains too slow and lacks efficient numerical methods.

Deterministic channel models based on ray tracing (Egea-Lopez et al. 2021; Vitucci et al. 2024; Xia et al. 2024; Zhang et al. 2024; Vaara et al. 2025) achieve high accuracy and capture a broad spectrum of channel characteristics, such as delay and angular spread, by exploiting the detailed 3D environment geometry and enabling calibration against measurement data. However, they depend on precise electromagnetic properties of building materials such as permittivity and conductivity, and inaccuracies in these properties can degrade the prediction quality. Moreover, the computational complexity of ray tracing often makes it impractical for rapid, large-scale coverage assessments in radio network planning and optimization. Mobile network operators still widely rely on empirical Hata-type models calibrated with the help of drive-test data and supplemented by diffraction loss in urban areas where building data is available (ElNashar 2014). For lower- and mid-band frequencies, both high spatial resolution and broad

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coverage—on the order of several tens of kilometers—are essential for cell overlap and interference analysis. Under these conditions, fast propagation models are needed not only for predictive accuracy but also for practical network planning. Preliminary interactive design on desktop workstation platforms requires prediction times on the order of minutes for coverage radii of about 1 km, enabling iterative refinement of site layouts and coverage parameters. In contrast, larger and more detailed final designs can be performed in batch mode on servers or computing clusters, where runtime constraints are less stringent.

Substantial research has been conducted over many years on raster-based analysis algorithms applied to radio or optical wave propagation. A broad range of these works deals with viewshed or line-of-sight visibility analysis (Wang and Dou 2024). The simplest method to calculate visibility around a site location is to follow all raster points and estimate visibility along each line between the antenna and the mobile receiver. In this way, the algorithm traverses every intermediate raster point between the transmitting and receiving antennas, which results in a computational complexity of $\mathcal{O}(N^3)$ (also called R3) (Franklin et al. 1994). This algorithm is slow, but accurate since it does not omit any obstacles along the radio wave path. Faster R2-type algorithms make a simplification by estimating obstacle heights at the edges of the analysis area and at the intersections of radial transmitter-receiver lines with the coordinate grid. Heights of the interior points are obtained at the closest propagation line intersections with the coordinate grid (Franklin et al. 1994). R2 algorithms run in $\mathcal{O}(N^2)$ time. There is an approximate RFVS algorithm (Franklin and Ray 1994), which also runs in $\mathcal{O}(N^2)$ time and analyzes obstacles between the transmitter and the receiver along radial rays, estimating the closest intersecting raster pixels. Kreveld (1996) proposed a sweep line approach, which runs in $\mathcal{O}(N \log N)$ time by calculating heights for raster pixels along a radial line from the cell center. In this case, raster pixels are arranged using balanced binary search trees (BST) for faster processing. Recently PDERL algorithm was proposed using reference lines in a proximity, direction and elevation (PDE) coordinate system, reaching an accuracy close to R3 (Wu et al. 2020; Cheng and Dou 2022). However, PDERL is more oriented to viewshed and line-of-sight analysis and less suitable for extensions to diffraction modeling using vector data. The widely used XDraw algorithm for raster-based calculations was first proposed by Franklin et al. (1994), later optimized and extended to different applications, including high-performance Apache Spark implementations (Zhu et al. 2019; Jianbo et al. 2019; Zhang et al. 2021; Dong and Zhang 2023). We will use the XDraw algorithm as the basis for building faster approximate methods for vector-based diffraction calculations.

Statistical propagation models are often insufficient for practical radio network coverage predictions, so realistic 3D city data is essential for site-specific modeling. In response, several 3D vector building datasets have recently become widely available, and the Open Geospatial Consortium's City Geography Markup Language (OGC CityGML) standard (Gröger et al. 2012) is leading the effort to unify this vector data. A good example is the 3D building model of New York City provided by NYC OpenData (2023), which has already been used for analyzing the importance of building data accuracy in radio coverage models (Simić et al. 2015). As shown in Beekhuizen et al. (2014), uncertainties in input building data are mostly responsible for the accuracy of radio-frequency electromagnetic fields (RF-EMF) exposure level predictions from mobile base stations, especially in non-line-of-sight conditions. Accuracy of 3D environment data is of primary importance also for diffraction, ray tracing, and machine learning-based radio propagation predictions (Xia et al. 2024; Zhang et al. 2024; Vaara et al. 2025; López-Ramírez et al. 2024). The proliferation of high-accuracy propagation prediction models is driven by the wide availability of open 3D building datasets which have appeared recently, including 3DCityDB (Yao et al. 2018), Paris-CARLA-3D (Deschaud et al. 2021), City3D (Huang et al. 2022), Building3D (Wang et al. 2023), BostonTwin (Testolina, Polese, Johari, and Melodia 2024), and 3D-GloBFP (Che et al. 2024), to name a few.

This work aims to develop a fast numerical diffraction algorithm tailored to high-accuracy 3D vector building data. By using vector-based inputs, it achieves greater precision in path loss prediction compared to traditional raster-based methods. In our model, diffraction loss is estimated according to the ITU-R P.526 recommendation (ITU-R 2019). The method is based on finding the clearance distance between the highest obstacle along the radio wave path and the radio beam center-line. The clearance is used for estimating the Fresnel diffraction integral, giving diffraction loss due to the obstructed path. This diffraction loss can be further combined with empirical, drive-test-calibrated propagation models such as Hata-type models.

The algorithm adapts the XDraw method to vectorized building data and runs in $\mathcal{O}(N^2)$ time by expanding concentric rings from the antenna across a uniform square grid and visiting each point exactly once. At each grid point, it performs a full geometric analysis to locate building intersections and then reuses diffraction results from the nearest points. Two approximation schemes speed this reuse:

- Max-Obst, or maximum obstacle method, which identifies the previously visited point with the highest diffraction loss, re-projects its obstructing feature onto the current point, and recalculates the loss;
- Max-Roof, or maximum roof method, which carries forward only the first few dominant roof planes from earlier points for diffraction computations.

The paper is organized as follows. Section 2 introduces Max-Obst and Max-Roof vector XDraw algorithms and explains their implementation principles. Section 3 gives information about the 3D building dataset used for testing XDraw diffraction algorithms. In Section 4, accuracy and time performance results are presented, followed by application of the proposed algorithms to the Internet of Things (IoT) network coverage analysis. Finally, conclusions are drawn.

2 | Methodology

In this section, we review the principles of XDraw algorithm and provide descriptions of vector-based XDraw algorithms for working with 3D building polygons. The idea of XDraw

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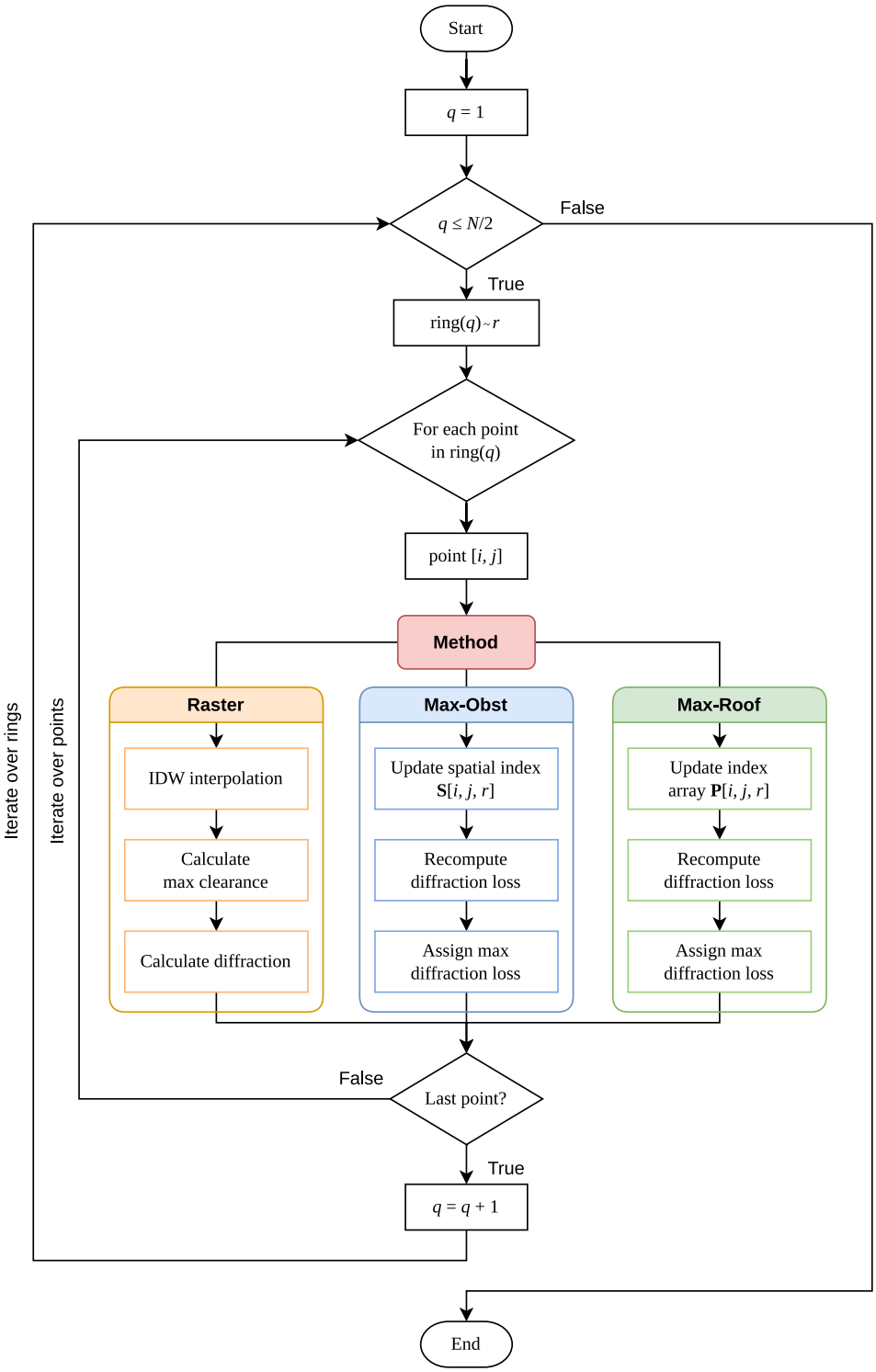


FIGURE 1 | Flowchart of the raster- and vector-based XDraw algorithms, highlighting the main iteration steps of spatial interpolation and diffraction analysis.

algorithm is to reuse previous calculations of diffraction along the line from the antenna to the receiver in order to save computation time. Square analysis area with N points in horizontal and vertical dimensions is divided into $N/2$ concentric square rings. The algorithm starts at base station antenna location at the center and proceeds toward edges of the area calculating diffraction loss for each point by iterating over rings. General flowchart of raster-based and vector-based algorithms is shown in Figure 1, illustrating two loops, the first along rings and the second along each ring. For each spatial index, the processing depends on whether the algorithm implementation is raster-based or vector-based. The pseudo code of raster XDraw algorithm is shown in Algorithm 1. Here two indexed vectors $\mathbf{H}[i,j,r]$ and $\mathbf{D}[i,j,r]$ are used to store maximum building height and distance from the antenna, respectively, at each grid location (i,j) for two consecutive rings

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ALGORITHM 1 | Raster-Based XDraw Diffraction Algorithm.

Input:
Building heights raster $\mathbf{B}$ of size $N \times N$, base station and mobile antenna heights h_{BS} and h_{M}, frequency f
Output:
Diffraction loss raster $\mathbf{L}$ of size $N \times N$
Initialize:
$\mathbf{H}[i,j,r] \leftarrow \emptyset, \quad \mathbf{D}[i,j,r] \leftarrow \emptyset, \quad i \in [1,N], \quad j \in [1,N], \quad r \in \{1,2\}$
for $q \leftarrow 1$ **to** $N/2$ **do**
  for each point$[i,j]$ **in** ring$[q]$ **do**
    $(i_1,j_1),(i_2,j_2) \leftarrow$ neighbor_idx(ring$[q-1]$) ▷ Indices of two neighbors from previous ring
    $\mathbf{H}[i,j,r] \leftarrow \mathrm{IDW}(\mathbf{H}[i_1,j_1,r-1], \mathbf{H}[i_2,j_2,r-1])$
    $\mathbf{D}[i,j,r] \leftarrow \mathrm{IDW}(\mathbf{D}[i_1,j_1,r-1], \mathbf{D}[i_2,j_2,r-1])$
    $\mathbf{H}[i,j,r] \leftarrow \mathbf{H}[i,j,p] \cup \{\mathbf{B}[i,j]\}$
    $\mathbf{D}[i,j,r] \leftarrow \mathbf{D}[i,j,p] \cup \{\mathrm{distance_from_BS}[i,j]\}$
    $\Delta h \leftarrow \max\{\mathrm{clearance}(\mathbf{H}[i,j,r], \mathbf{D}[i,j,r], h_{\mathrm{BS}}, h_{\mathrm{M}})\}$
    $\nu \leftarrow$ diffraction_parameter$(\Delta h, d_1, d_2, f)$ ▷ d_1, d_2 are distances from max obstacle
    $\mathbf{L}[i,j] \leftarrow$ diffraction_loss(ν)
  end for
end for
return L

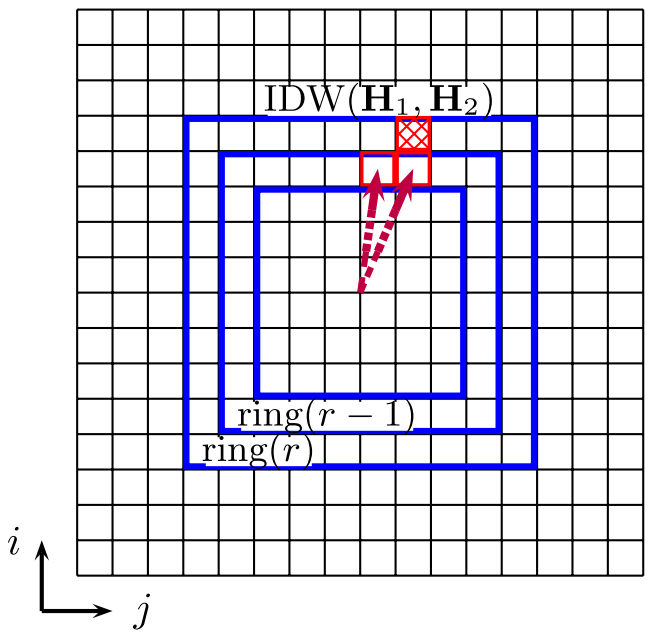


FIGURE 2 | IDW interpolation in the XDraw algorithm. The hatched red rectangle indicates the current spatial index, while the empty red rectangles represent previously visited spatial indices to be interpolated. The red arrows show the iteration progressing from the center of the analysis area outward in square rings.

with indices $r \in 1, 2$. At each iteration over rings, these vectors at the current ring r are assigned corresponding values from the previous ring $r-1$ by applying inverse distance weighted (IDW) interpolation for the two vectors from previous neighbors (Figure 2). Then height **H** and distance **D** vectors are extended by adding building height and distance of the current grid location (i,j). After that, radio path clearance is estimated by selecting maximum height and corresponding height from **H** and **D** vectors resulting in the lowest path clearance due to all buildings in between the base station antenna and the mobile receiver. Having found minimum path clearance, the diffraction loss is estimated following ITU-R P.526–15 recommendation (ITU-R 2019). This procedure ensures that each grid location is visited only once when calculating minimum clearance and maximum diffraction loss along the radio path.

In order to apply the same XDraw principle to vector building data consisting of roof and wall polygons, we need to replace IDW interpolation used for averaging raster data from two previous grid points with another approximation method working directly with 3D geometry objects. For 3D vector building data, diffraction can be calculated by the same ITU-R P.526 model after estimating the path clearance from the exact radio line and building geometry. The clearance in this case is the shortest distance between the radio line and the nearest building. The algorithm should find all intersecting roof and wall polygons of buildings with the radio line and calculate distances between the line and edges of each polygon. For large numbers of building polygons in urban scenarios, some approximation is needed to avoid full analysis of all the buildings between the transmitter and the receiver. A potential solution would be to select dominant buildings or roofs along the radio line and limit diffraction estimation to these geometries in subsequent XDraw ring iterations. Since usually in the 3D building data structure the building objects are not well separated and there are closely collocated building blocks in high-density streets, a proper method is needed to identify and store obstacles along the radio line during XDraw iterations. One possible way

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ALGORITHM 2 | Vector-Based XDraw Diffraction Max-Obst Algorithm.

Input:
Polygons dataset $P = \{p_1, \ldots, p_n\}$, spatial index S of polygons
$p \in P$ intersecting grid index sid_{ij}, parameter $k_{\max} \geq 1$
Output:
Diffraction loss raster $\mathbf{L}$ of size $N \times N$
Initialize:
Define array $\mathbf{S}[i, j, r]$ with entries $\{(sid_1, L_1), \ldots\}$
Initialize $\mathbf{S}[i, j, r] \leftarrow \emptyset$ for all $i, j \in [1, N]$, $r \in \{1, 2\}$
for $q \leftarrow 1$ **to** $N/2$ **do**
  for each point$[i, j]$ in ring$[q]$ **do**
    $(i_1, j_1), (i_2, j_2) \leftarrow$ neighbor_idx(ring$[q-1]$) ▷ Indices of two neighbors from previous ring
    $(sid_1, L_1) \leftarrow \mathbf{S}[i_1, j_1, r-1]$; $(sid_2, L_2) \leftarrow \mathbf{S}[i_2, j_2, r-1]$
    $T \leftarrow \emptyset$
    if $k = 1$ **then**
      let $m^* = \arg\max_{m \in \{1,2\}} L_m$, set $T \leftarrow T \cup \{sid_{m^*}\}$
    else
      $T \leftarrow T \cup \{sid_1, sid_2\}$
    end if
    $T \leftarrow T \cup \{sid_{ij}\}$ ▷ Spatial index of current grid point $[i, j]$
    Recompute diffraction losses for all $p \in T$, updating each L_m
    let $m^* = \arg\max_{m \in \{1, \ldots, |T|\}} L_m$
    $\mathbf{S}[i, j, r] \leftarrow \{(sid_{m^*}, L_{m^*})\}$
    $\mathbf{L}[i, j] \leftarrow L_{m^*}$
  end for
end for
return L

would be to divide the analysis area into an equidistant spatial grid with the required resolution and collect intersecting building polygons for each spatial grid pixel. Then XDraw algorithm can be applied over all pixels of the spatial grid by analyzing polygons at each pixel. In this work, we propose two approximation strategies for limiting building geometry analysis along the path: maximum obstacle (Max-Obst) and maximum roof (Max-Roof) methods. In Max-Obst method, building polygons in each grid point are analyzed only once when traversing the current XDraw ring and diffraction loss is stored together with spatial index in a data array which is extended for each ring iteration. This makes it possible to sort the data array by diffraction loss, thus determining the dominant grid point along the radio line and including building polygons from previously visited grid points into the diffraction analysis. In an alternative method, Max-Roof, roof polygons with dominant diffraction are stored in a data array for subsequent use in outer XDraw rings. When using only roofs instead of all polygons per pixel, more polygons from different spatial grid locations can be stored and analyzed. In both methods, only partial building geometry is analyzed, thus sacrificing accuracy for performance.

In more detail, vector-based Max-Obst and Max-Roof methods are explained in Algorithms 2 and 3, respectively. Both methods accept as input the building geometry data in the form of a polygon dataset P and a spatial index dataset S over a grid of size $N \times N$. Dataset S contains polygons intersecting spatial grid pixel with index sid_{ij}. Max-Obst method starts by building indexed array $\mathbf{S}[i,j,r]$ containing pairs of spatial grid index and calculated diffraction loss $\{(sid_n, L_n)\}$. Each iteration over XDraw rings starts by determining previous neighbors from the last inner ring $r-1$ and collecting $\mathbf{S}[i,j,r]$ from the neighbors into an empty T array. If the maximum number of obstacles $k_{max} = 1$, only pixel with the maximum diffraction loss is taken, otherwise spatial indices of both neighbors are collected. Building geometry from the current grid pixel sid_{ij} is also added to list T, which is used to calculate diffraction loss. In this way, a maximum of three grid pixels are analyzed and the highest diffraction loss is used as value for the current (i,j) pixel of the output diffraction raster $\mathbf{L}[i,j]$. At the same time, current pixel index and calculated diffraction value are appended to array $\mathbf{S}[i,j,r]$.

The Max-Roof diffraction algorithm works in a similar manner using an indexed array $\mathbf{P}[i,j,r]$ containing pairs of building polygons and calculated diffraction loss $\{(p_n, L_n)\}$. Each ring r iteration selects the $\mathbf{P}[i_{\max}, j_{\max}, r-1]$ array from the previous ring $r-1$ with the greater diffraction loss from two neighbors. The array is sorted according to diffraction loss and the first $k_{\max}$ polygons with dominant diffraction are selected into $P_{k_{\max}}$ which are used to recalculate diffraction loss for the current pixel geometry while updating loss information in $\mathbf{P}[i,j,r]$ array for the next iterations. After finding maximum diffraction loss $L_{\max}$ from the recalculated polygons, its value is assigned to the current pixel of the diffraction raster $\mathbf{L}[i,j]$.

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ALGORITHM 3 | Vector-Based XDraw Diffraction Max-Roof Algorithm.

Input:
  Polygons dataset $P = \{p_1, \ldots, p_n\}$, spatial index S of polygons
  $p \in P$ intersecting grid index sid_{ij}, parameter $k_{\max} \geq 1$
Output:
  Diffraction loss raster $\mathbf{L}$ of size $N \times N$
Initialize:
  Define array $\mathbf{P}[i, j, r]$ with entries $\{(p_1, L_1), \ldots\}$
  Initialize $\mathbf{P}[i, j, r] \leftarrow \emptyset$ for all $i, j \in [1, N]$, $r \in \{1, 2\}$
for $q \leftarrow 1$ **to** $N/2$ **do**
  for each point$[i, j]$ in ring$[q]$ **do**
    $(i_1, j_1), (i_2, j_2) \leftarrow$ neighbor_idx(ring$[q-1]$)
    $(p_1, L_1) \leftarrow \mathbf{P}[i_1, j_1, r-1]$; $(p_2, L_2) \leftarrow \mathbf{P}[i_2, j_2, r-1]$
    let $m^* = \arg\max_{m \in \{1,2\}} L_m$, and (i^*, j^*) the coords of p_{m^*}
    $\mathbf{P}[i, j, r] \leftarrow \mathbf{P}[i^*, j^*, r-1]$
    Compute diffraction losses for all $p \in sid_{ij}$, pick top $k_{\max}$ by loss
    Add those $\{(p_s, L_s)\}$ to $\mathbf{P}[i, j, r]$
    Recompute losses for these $k_{\max}$ entries
    $\mathbf{L}[i, j] \leftarrow \max_s L_s$
  end for
end for
return L

3 | Buildings Dataset

In this work, 3D building data (3D BAG) of the Netherlands reconstructed from two-dimensional building footprints and LiDAR data is used (Peters et al. 2022). This is the second version of the building dataset which covers the whole area of the Netherlands with about 10 million buildings and is derived from data with an accuracy of 30 cm. The geometry of the buildings is stored in the CityJSON format with the highest level of detail, LOD 2.2. It allows detailed visualization of the shapes of building roofs and facades. An example of the 3D building data is shown using polygons representing walls and roofs in Figure 3, illustrating the Royal Palace in the center of Amsterdam.

The geometry of the buildings is stored as polygons representing the walls and roofs of buildings. The data is organized in tiles whose size depends on the building density in the area. Three building datasets of different sizes were assembled for diffraction algorithm testing. The first dataset, T1, was created from a single tile #7190 with the total area extended around the buildings to 2 km × 2 km, the second dataset, called T10, contains 10 tiles and spans an area of 3 km × 3 km, and the third dataset, called T300, comprises 293 tiles and spans an area of about 30 km by 20 km. Table 1 summarizes the datasets, listing the numbers of buildings, roof polygons, and wall polygons. The layout of the building footprints inside the datasets is shown in Figure 4.

The building walls and roofs are defined by polygons, each comprising a set of 3D point coordinates. When cleaning the data, incomplete polygons with less than three points were discarded. In addition, polygon simplification was applied to the building

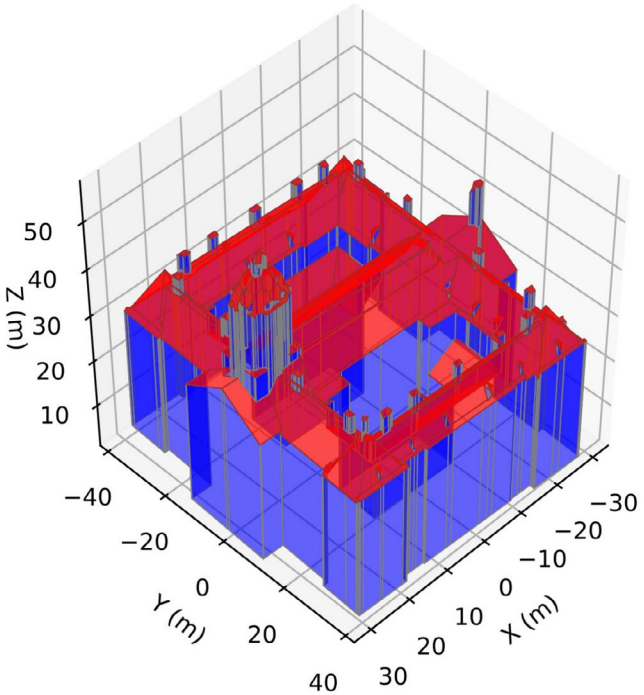


FIGURE 3 | Example of building data with level of detail LOD 2.2 from 3D BAG open data of the Netherlands (Peters et al. 2022).

polygons to reduce the data size by omitting vertices along the polygon edge with an internal angle between neighboring vertices greater than 178°. In this way, an angular error of 2° is introduced by replacing two edges forming a nearly straight angle with a single edge.

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4 | Results

4.1 | Accuracy and Performance of Diffraction Algorithms

Approximate diffraction prediction methods were tested on Amsterdam city datasets using two computer configurations with Intel CPUs: Xeon E5-2670 and Core i9-10900, the main parameters of which are listed in Table 2. These platforms are representative of workstation-class systems suitable for interactive large-scale geospatial data processing. Algorithms were implemented using Python 3.12 numerical libraries with Numba (Lam et al. 2015) JIT-compiled functions on the Linux Ubuntu 24.10 operating system. For accuracy and performance testing, single processing and 8 processes in parallel were used on a 1 m resolution grid. Diffraction model accuracy was estimated with respect to the geometrically exact diffraction algorithm which analyzes radio line intersections with all building polygons between the transmitter and the receiver. Therefore, the exact model is slow and runs in $\mathcal{O}(N^3)$ time. Diffraction loss prediction results for base station height 45 m, mobile height 2 m, operating frequency 900 MHz and cell radius 600 m are shown in Figure 5. Here, the left image shows the exact model results, the center image—XDraw Max-Roof with $k = 3$ approximation, and the right image displays the difference. As shown by the difference raster, the highest errors appear at distances farther away from buildings, where the Max-Roof model loses track of obstacles through longer loops over rings in the XDraw algorithm. It also shows that vector-based XDraw algorithms underestimate diffraction compared to geometrically exact predictions.

The accuracy of different XDraw approximations is compared in Table 3, where the mean absolute error $\mathrm{MAE} = 1/n \left\| \mathbf{L} - \hat{\mathbf{L}} \right\|_1$ and the root mean square error $\mathrm{RMSE} = 1/\sqrt{n} \left\| \mathbf{L} - \hat{\mathbf{L}} \right\|_2$ are estimated between the exact diffraction loss raster $\mathbf{L}$ and the approximate raster $\hat{\mathbf{L}}$ using L1 and L2 norms, respectively. For comparison, we also include raster-based model results: 'Raster R3' is the exact $\mathcal{O}(N^3)$ algorithm that computes diffraction over the building height raster sampled at a 1 m resolution; 'Raster R2' is the raster–based XDraw algorithm; and the last two raster algorithms marked with '× 2' denote versions resampled at twice the resolution (0.5 m). In most cases, the accuracy of raster-based models is worse than that of vector-based models, comparable only for vector models with a single obstacle $k = 1$ included in the analysis. Even upsampling of the building height data does not yield significant improvements for raster algorithms. The best performing model is the vector XDraw Max-Roof algorithm; however, its accuracy settles around RMSE of 6.8 dB with $k = 3$ obstacles and does not improve significantly when increasing k. Vector XDraw Max-Obst model reaches RMSE of 8.7 dB with $k = 2$, which is better than that of raster-based algorithms; however, a higher number of obstacles for this model was not implemented due to slower performance compared to the Max-Roof model. For vector-based XDraw models, RMSE errors are significantly higher than MAE, indicating the dominance of larger diffraction calculation errors, mostly due to missed

TABLE 1 | Building datasets of Amsterdam city used in diffraction calculation experiments.

	T1	T10	T300
Number of tiles	1	10	293
Number of buildings	1110	13,141	357,792
Number of roofs	10,240	99,347	1,123,381
Number of walls	98,916	953,702	11,385,358
Area (km^2)	4	9	595

TABLE 2 | Computer configurations used for testing diffraction algorithms.

	Intel Xeon E5-2670	Intel Core i9-10900
CPU clockspeed	2.60 GHz	2.80 GHz
Number of physical cores	8	10
Number of threads	16	20
Number of CPUs per computer	2	1
Installed RAM	128 GB	64 GB

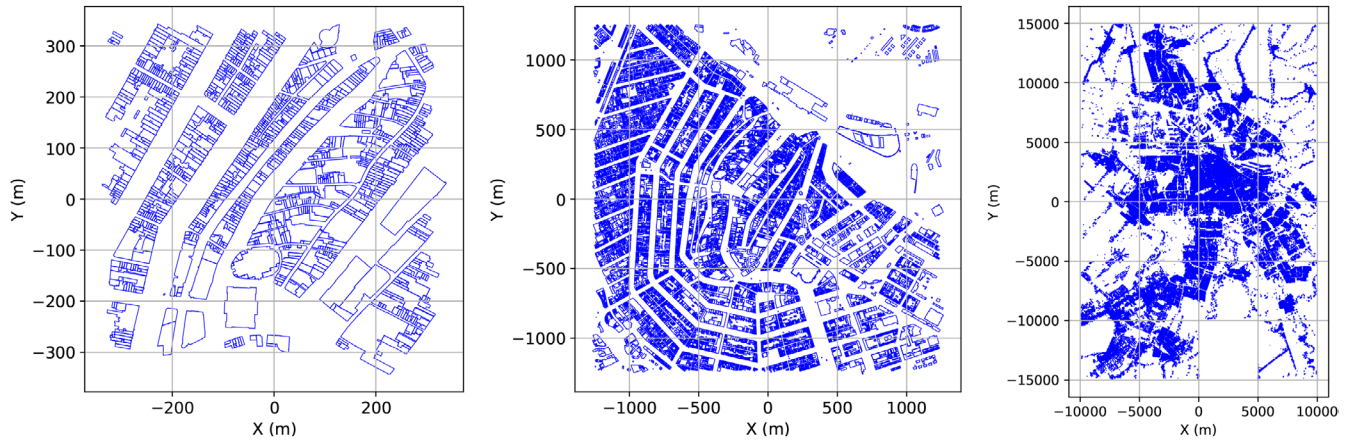


FIGURE 4 | Building footprints of T1 (left), T10 (center), and T300 (right) datasets (Peters et al. 2022).

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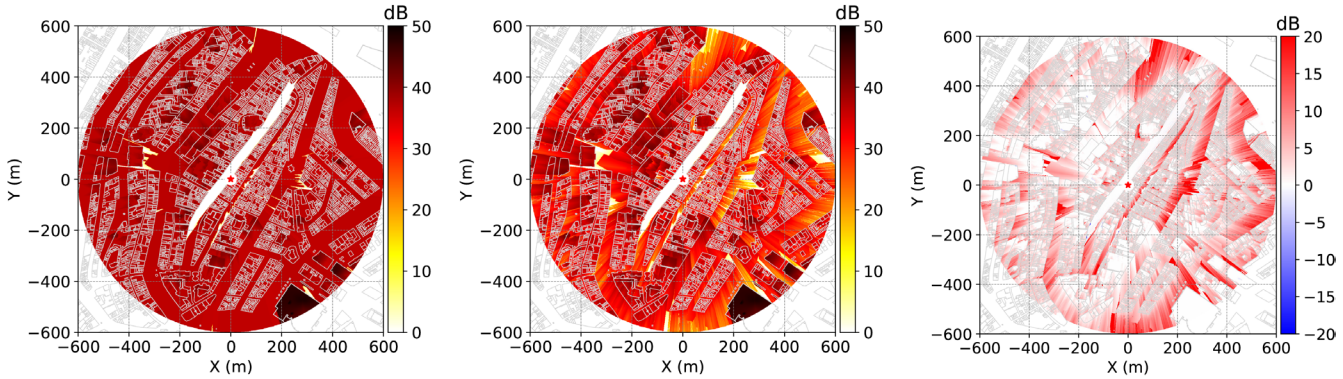


FIGURE 5 | Diffraction loss using exact method (left), approximate method Max-Roof k = 3 (center), and the difference (right).

TABLE 3 | MAE and RMSE error estimates between diffraction rasters calculated using the exact vector method and approximate vector- and raster-based methods.

Method	MAE (dB)	RMSE (dB)
Max-Roof, $k = 20$	4.2	6.7
Max-Roof, $k = 5$	4.3	6.8
Max-Roof, $k = 3$	4.3	6.8
Max-Roof, $k = 2$	4.5	7.3
Max-Roof, $k = 1$	8.0	14.1
Max-Obst, $k = 2$	4.7	8.7
Max-Obst, $k = 1$	5.9	11.1
Raster R3	8.1	9.5
Raster R2	9.0	10.9
Raster R3 × 2	6.4	8.3
Raster R2 × 2	7.2	9.6

obstacles further away from the buildings as shown in Figure 5. The distribution of diffraction error size for XDraw algorithms is shown in Figure 6 (left). Here, the cumulative distribution function (CDF) shows that the most accurate Max-Roof algorithm predicts 90% of the area with an error of less than 10 dB.

The time performance of vector XDraw algorithms is shown in Figure 6 (right) for cells with radii in the range from 150 to 1500 m calculated using 8 processes in parallel on Core i9-10900 CPU. The performance of Max-Roof algorithm is better than that of Max-Obst algorithm, suggesting Max-Roof algorithm to be the best both from the accuracy and the performance point of view. To better optimize the performance of Max-Roof algorithm, different parallelization methods for Python code were tested. When using parallel processing in Python, the performance is influenced by Python's global interpreter lock (GIL). Parallel processes can be created using the 'multiprocessing' module when each new process has its own Python interpreter and a separate GIL allowing it to run independently within its own area of memory. Python can also run parallel threads using the 'threading' module, in which case all threads share the same memory area and data. There are also other options for parallelization in Python, like the Dask library (Rocklin 2015), especially effective for larger datasets, and Numba's (Lam et al. 2015) for loop parallelization using the LLVM compiler library. XDraw algorithms can easily be parallelized by dividing the analysis area into multiple symmetrical radial segments. In this case, we used eight parallel processes for Max-Roof $k = 3$ diffraction prediction using vector data, and the performance results for different parallelization methods are shown in Figure 7. The difference between various parallelization methods is more evident for the smaller dataset T1 for both tested CPUs, with the 'multiprocessing' library giving the shortest calculation durations. A minor enhancement in the processing performance of 'multiprocessing' is also visible for the large T10 dataset, although in this case diffraction calculation durations are longer and the difference between parallelization methods is not significant. The time performance of parallel execution is also compared with the single-process (linear) calculation and the exact vector method, which is several orders of magnitude slower than XDraw algorithms. The timing results confirm the expected scaling behavior: the exact method exhibits approximately cubic growth, whereas the vector XDraw approximations scale approximately quadratically with analysis radius. For dataset T1, diffraction calculations with a radius of 1 km remain within 10 min on both CPUs. For the larger T10 dataset, the same time budget allows a radius of only about 500 m, while the runtime increases to slightly above 1 h for a radius of 1.5 km on the better-performing Core i9-10900 platform. Such computational performance is sufficient for interactive preliminary cellular network design when cell radius is limited to about 1 km. The speed of raster-based XDraw diffraction methods is not studied here, but from several implementations we tested, they are faster by 50–100 times compared to the vector algorithms for the same area size.

4.2 | Applications to IoT Network Coverage Estimation

To illustrate the influence of vector-based diffraction prediction on wireless network coverage results, we constructed two layouts of a cellular network with 7 and 25 base stations in the city of Amsterdam, and compared coverage results obtained using raster- and vector-based XDraw diffraction algorithms. First, two cellular configurations with regular hexagonal geometry were constructed, consisting of 7 and 25 omni-directional sites, as shown in Figure 8 (left). The inter-site distances (ISDs) for these configurations are 4 and 2 km, respectively. After creating regular configurations, base station locations were adjusted

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taking into account real base station positions available from Antennebureau (Antenna Office of the Netherlands, https://www.antennebureau.nl/). This antenna mast dataset also contains antenna heights above ground for GSM, UMTS, LTE and 5G NR installations. For selected cellular sites, antenna heights vary in the range of 20–50 m. One hexagonal cluster of seven sites was matched to existing cellular towers as shown in Figure 8 (center). Another cellular plan of six-site clusters consisting of a total of 31 sites was also adjusted to existing antenna locations, reusing seven sites from the first configuration with an ISD of 4 km and selecting 18 new positions from the antenna mast dataset, resulting in a total of 25 base stations. Six sites from the initial regular configuration were omitted due to an insufficient number of existing masts in parts of the network covering less dense areas. Both configurations of 7 sites with a maximum cell radius of 3 km and 25 sites with a maximum cell radius of 2 km are capable of ensuring continuous coverage of the area.

The created cellular configurations were used to estimate IoT network coverage predicted using standard raster-based XDraw diffraction algorithm and vector-based XDraw Max-Roof diffraction method with the first $k = 3$ dominant roofs. Total path loss was estimated using a combined model including free space path loss for line-of-sight (LOS) areas and Hata path loss plus diffraction for non-line-of-sight (NLOS) areas. Hata loss was calculated using Hata 9999 model or Ericsson implementation (ElNashar 2014) with parameters $a_0 = 36.2$, $a_1 = 30.2$, $a_2 = -12$ and $a_3 = 0.1$ adjusted for the urban environment. For areas covered by buildings, an additional building penetration loss of 20 dB was added.

For coverage predictions, the case of a typical narrowband IoT (NB-IoT) uplink scenario was selected with a transmit power of 23 dBm, a maximum coupling loss (MCL) of 164 dB and a receiver sensitivity of −141 dBm (Liberg et al. 2019). The widely

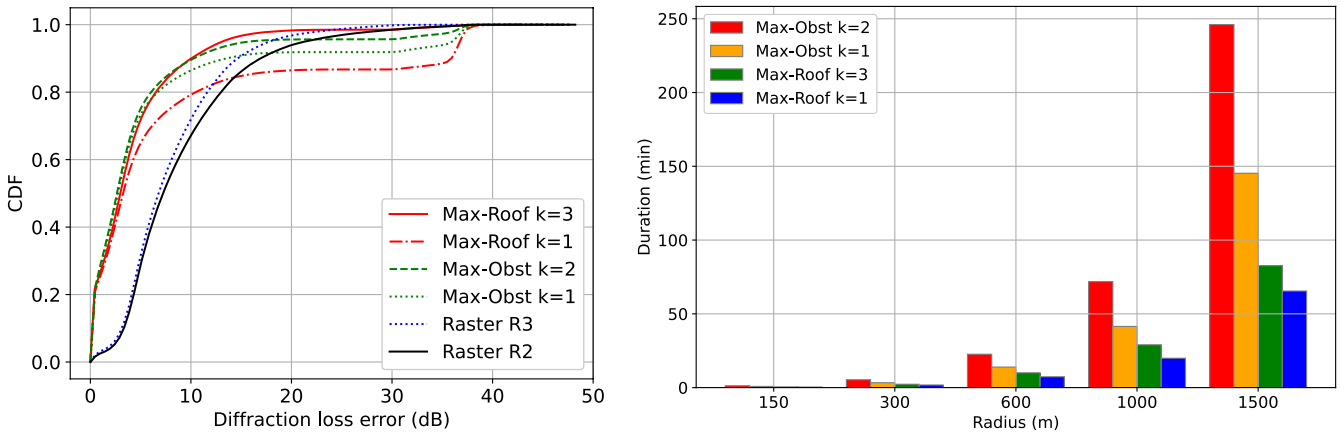


FIGURE 6 | Cumulative distribution function (CDF) of diffraction prediction error (left) and calculation duration (right) for various approximation methods for T10 dataset and Core i9-10900 CPU.

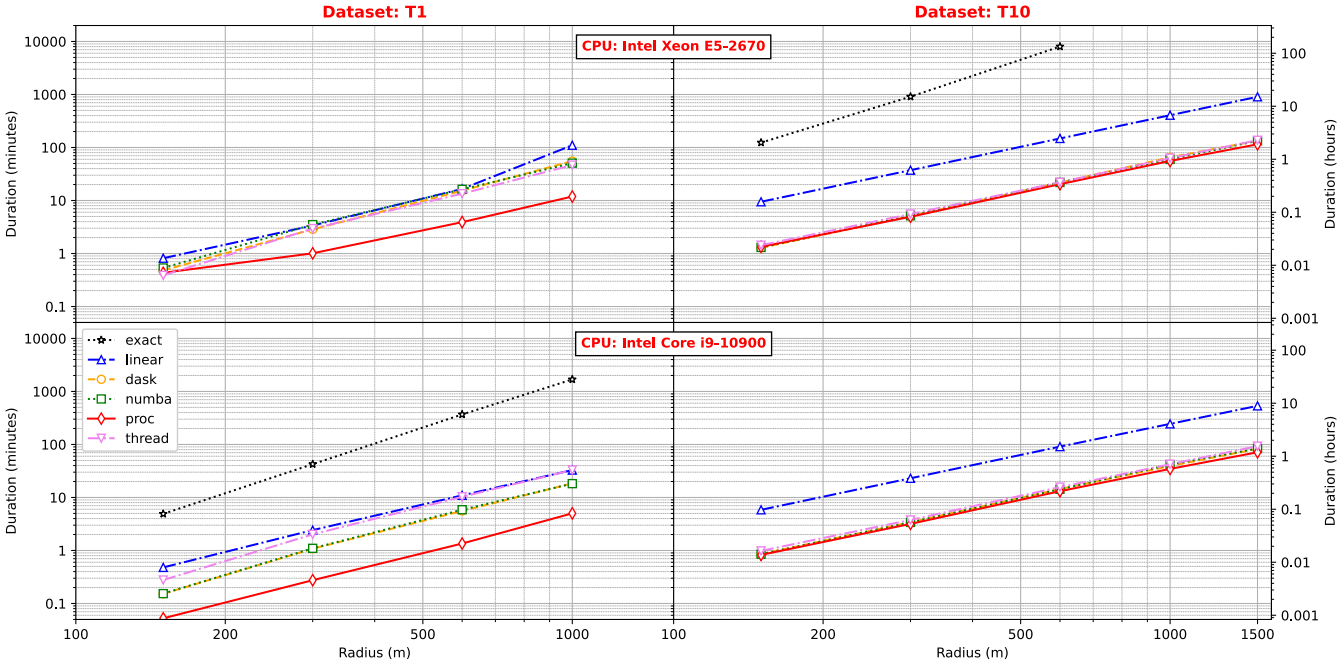


FIGURE 7 | Time performance characteristics of vector XDraw Max-Roof $_{k=3}$ algorithm with different parallelization methods using 8 processes.

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available 800 MHz frequency band was used for calculations and the mobile antenna height was set to 2 m. For simplicity, terrain elevation was not included in the analysis, considering as obstacles only building geometry. The whole analysis area was sampled by a 1 m resolution grid and the received signal level (RSL) was calculated for the 7-site and the 25-site clusters. For the 7-site cluster, uplink coverage predictions calculated using vector XDraw Max-Roof model are shown in Figure 9: the RSL is shown on the left and the omni-directional cell areas on the right. Higher levels of the received signal in the lower part of the network indicate free space propagation due to open areas, while the rest of the network is predominantly covered by NLOS propagation.

The comparison of statistical distributions of RSL values over the network area calculated using vector- and raster-based XDraw diffraction algorithms is shown in Figure 10 (left). Here, the most accurate vector-based Max-Roof $k = 3$ algorithm is applied. Higher RSL values are affected mostly by line-of-sight propagation, therefore there is no significant difference between vector and raster diffraction algorithms. However, the lower part of the RSL CDF distributions shows a divergence of about 12 dB. The difference between vector and raster XDraw diffraction algorithms also leads to an offset in line-of-sight probability dependence on distance shown in Figure 10 (right). In this case, XDraw diffraction algorithm underestimates the line-of-sight probability by up to about 2%.

5 | Conclusions

The results demonstrate that XDraw diffraction approximations can be effectively applied to 3D building data. The Max-Roof XDraw variant achieves its lowest approximation error when at least $k = 3$ obstacles are included; adding more obstacles beyond this point yields negligible accuracy gains. Although vector-based XDraw implementations are slower than their raster-based counterparts, they can leverage multi-process parallelization to boost overall computation speed. The best parallelization results in Python are achieved using process-based 'multiprocessing' libraries. However, even when JIT-compiled with Numba, Python code is not optimal, and code recompilation using C/C++ libraries should provide better computational performance. The presented results show the advantage of vector XDraw algorithms, enabling higher accuracy due to the lower loss of geometry information compared with raster XDraw algorithms when R2-type approximations are applied to speed up the processing of building height datasets. The accuracy of vector-based XDraw

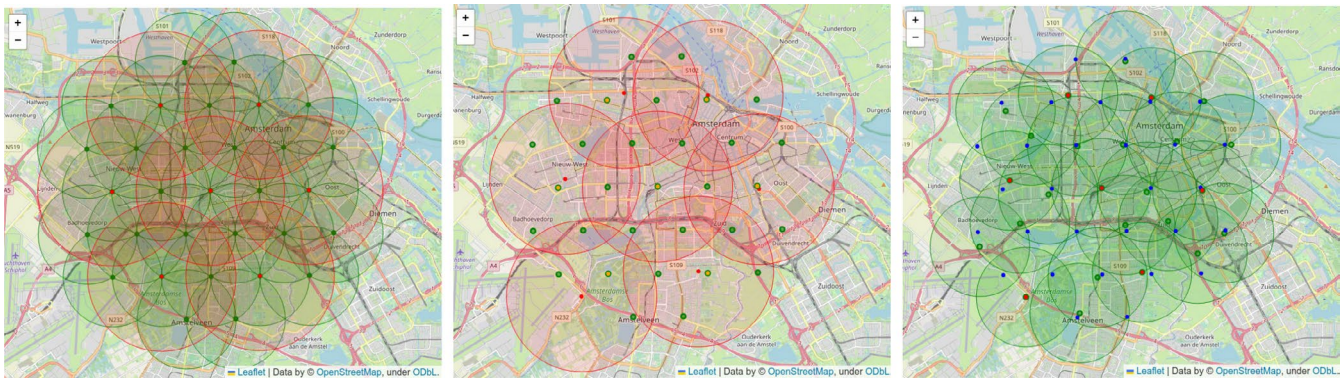


FIGURE 8 | Hexagonal omni-cell configurations of 7 sites (red color) and 31 sites (green color) shown on the left, and the same cell configurations adjusted to existing antenna tower locations with 7 sites (center) and 25 sites (right). Basemap data provided by OpenStreetMap (2026).

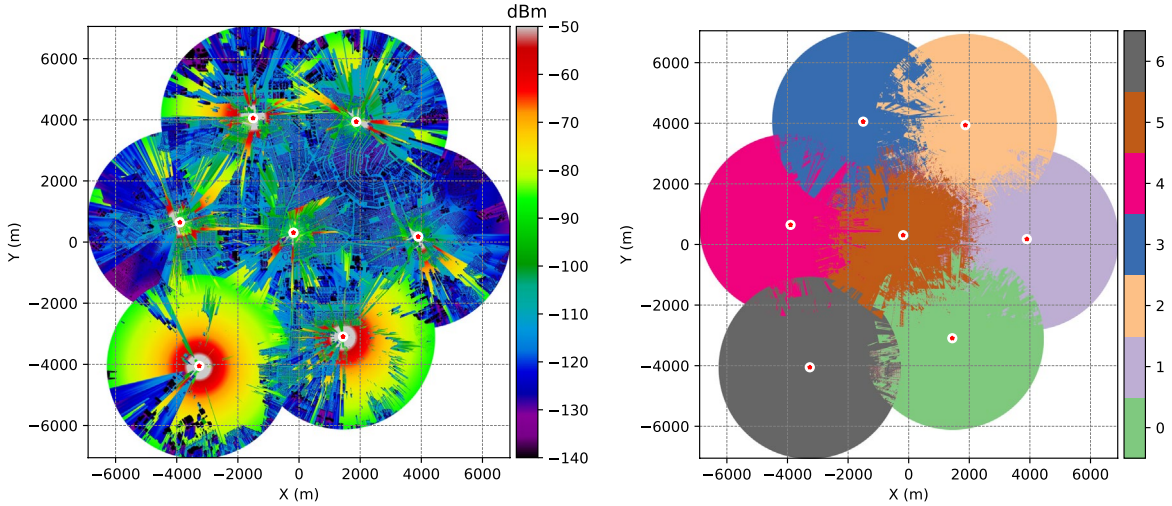


FIGURE 9 | NB-IoT uplink coverage results: Received signal level (left) and true cell areas (right) for 7 site cluster predicted using XDraw Max-Roof model with $k = 3$.

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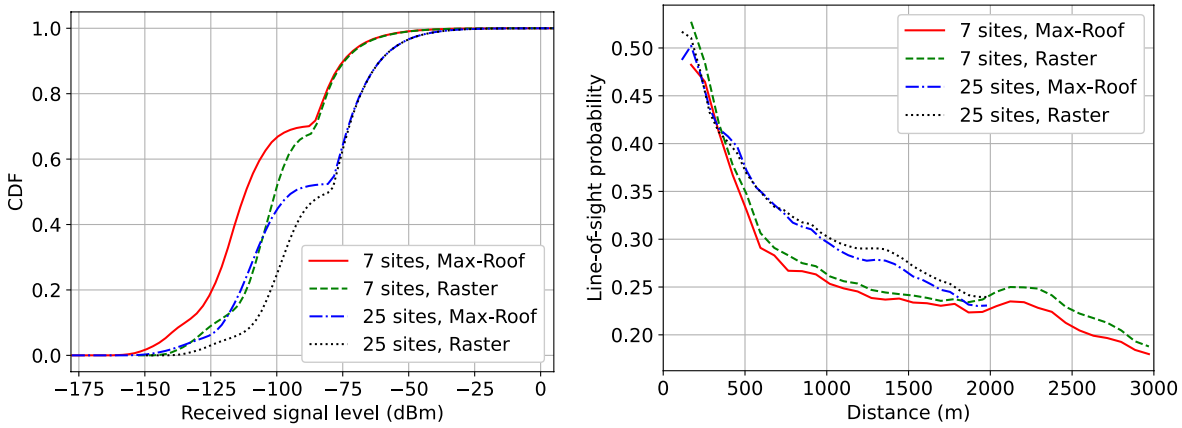


FIGURE 10 | Statistical distribution of received signal level (left) and line-of-sight probability (right) calculated using vector-based Max-Roof and raster-based XDraw diffraction algorithms.

algorithms can be further increased since they enable the use of not only the ITU-R P.526 diffraction model in the vertical plane, but also horizontal diffraction models and other advanced diffraction algorithms.

As 3D vector building data becomes increasingly available from LiDAR measurements, its use in radio signal coverage prediction enables resolutions on the order of 1 m or even sub-meter scales. This capability is particularly important for the adoption of new radio spectrum in centimeter- and millimeter-wave bands, where obstacles with dimensions comparable to the wavelength can become critical for propagation. Such frequency ranges are expected to support short-range, high-capacity links in dense urban environments to meet the traffic demands of future wireless networks.

From a practical implementation perspective, vector-based XDraw algorithms can be integrated into propagation modeling workflows in a manner similar to traditional raster-based approaches. The proposed algorithms follow the same ring-based traversal of the analysis grid as the original XDraw method, which simplifies their incorporation into existing coverage prediction frameworks. The computational procedure is also well suited for parallel execution since the analysis area can be partitioned radially into independent spatial sectors. This enables straightforward parallelization across multiple processes or computing nodes, allowing efficient processing of large urban areas. In addition, the achieved runtime on the order of 10 min for a coverage radius of about 1 km indicates that the proposed approach is practical for interactive preliminary network design on workstation-class platforms. Another advantage is the direct use of vector building datasets without rasterization, thereby avoiding additional preprocessing steps while preserving geometric fidelity. The approach is also compatible with standard GIS data formats used in modern 3D city models, such as CityGML and CityJSON, thereby facilitating integration with existing urban data infrastructures and propagation modeling platforms.

Future work may focus on further improving computational performance and extending the modeling framework. Potential directions include optimized implementations in compiled languages such as C/C++, development of more sophisticated spatial indexing structures for faster geometric queries, GPU acceleration of geometric transformations, and incorporation of additional propagation mechanisms such as horizontal diffraction and multi-edge diffraction models. In addition, combining vector-based diffraction analysis with hybrid deterministic-empirical models may further improve prediction accuracy for large-scale wireless network planning and optimization.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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